Proposition 1 (Uniform upper bound for $\mathcal{L}^{\text{adv}}(\theta)$)**.** *For the AT loss function $\mathcal{L}^{\text{adv}}(\theta)$ defined in Eq. (8) and the surrogate AT loss function $\tilde{\mathcal{L}}^{\text{adv}}(\theta)$ defined in Eq. (9), for any model parameter $\theta := (W^V, W^{KQ})$ of the LSA model $f_{\text{LSA},\theta}$, we uniformly have that: $\mathcal{L}^{\text{adv}}(\theta) \leq \tilde{\mathcal{L}}^{\text{adv}}(\theta)$.*

This result indicates that when we are training the LSA model via solving the surrogate AT problem Eq. (9), we are also reducing the model training loss in the original AT problem Eq. (8). Thus, solving the surrogate AT problem will also intuitively improve the robustness of the model.

Based on our previous analysis, we now turn to study the training dynamics of surrogate AT defined in Eq. (9). To better describe our results, we define two functions $\Gamma(\cdot) : \mathbb{N} \to \mathbb{R}^{d\times d}$ and $\psi(\cdot) : \mathbb{N} \to \mathbb{R}$, both of which depend on the adversarial suffix length M, as follows,

$$\Gamma(M) := \frac{N+M+1}{N+M}\Lambda + \frac{\text{Tr}(\Lambda)}{N+M} I_d \in \mathbb{R}^{d\times d}, \quad \psi(M) := \frac{M^2 \text{Tr}(\Lambda)}{(N+M)^2} \in \mathbb{R}, \tag{10}$$

where N is the prompt length of the original ICL input E_τ (see Eq. (4)) and Λ is the covariance matrix of in-context linear regression samples. The closed-form surrogate AT dynamics of the LSA model $f_{\text{LSA},\theta}$ is then given in the following Theorem 1 (see Appendix A.3 for the proof).

Theorem 1 (Closed-form Surrogate AT Dynamics)**.** *Suppose Assumption 1 holds and $f_{\text{LSA},\theta}$ is trained from the surrogate AT problem defined in Eq. (9) with continuous gradient flow. When the σ in Assumption 1 satisfies $\sigma < \sqrt{\frac{2}{d\cdot\|(\Gamma(M_{\text{train}})\Lambda+\epsilon^2\psi(M_{\text{train}})I_d)\Lambda^{-1}\|_2}}$, after training for infinite long time, the model parameter θ will converge to $\theta_*(M_{\text{train}}) := (W_*^V(M_{\text{train}}), W_*^{KQ}(M_{\text{train}}))$, satisfying: $w_{*,12}^{KQ} = w_{*,21}^{KQ} = w_{*,12}^{V} = w_{*,21}^{V} = 0_{d\times 1}$, $w_{*,22}^{KQ} = 0$, $W_{*,11}^{V} = 0_{d\times d}$, and*

$$w_{*,22}^{V} W_{*,11}^{KQ} = \Big(\Gamma(M_{\text{train}})\Lambda + \epsilon^2\psi(M_{\text{train}})I_d\Big)^{-1}\Lambda.$$

Remark 1. *When the l_2-norm adversarial perturbation radius ϵ is zero, the closed-form AT solution θ_* derived in Theorem 1 degenerates to that obtained without AT (see Theorem 4.1 in [69]). Thus, a sufficient large adversarial perturbation ϵ is a key to helping the LSA model $f_{LSA,\theta}$ obtain significant adversarial robustness. This will be further justified in the next section.*

4.4 Robust generalization upper-bound

With the closed-form AT solution $\theta_*(M_{\text{train}})$ in Theorem 1, we now analyze the robustness of the trained LSA model. All proofs in this section are presented in Appendix A.4. We study how a LSA model adversarially trained under a fixed adversarial suffix length M_{train} can defend against the ICL adversarial attack with a different adversarial suffix length M_{test}. That is, we aim to analyze the magnitude of the robust generalization error $\mathcal{R}^{\text{adv}}(\theta_*(M_{\text{train}}), M_{\text{test}})$ for the converged robust model parameter $\theta_*(M_{\text{train}})$. Here, we prove an upper-bound for it in the following theorem.

Theorem 2 (Surrogate AT Robust Generalization Bound)**.** *Suppose all conditions in Theorem 1 hold and $\theta_*(M_{\text{train}})$ is the surrogate AT solution in Theorem 1. We have*

$$\mathcal{R}^{\text{adv}}(\theta_*(M_{\text{train}}), M_{\text{test}}) \leq 2\text{Tr}\Big[\Lambda^3\Big(\Gamma_{\text{test}}\Lambda + \epsilon^2\psi_{\text{test}}I_d\Big)\Big(\Gamma_{\text{train}}\Lambda + \epsilon^2\psi_{\text{train}}I_d\Big)^{-2} + \Lambda\Big],$$

where M_{train} is the adversarial suffix length in the ICL adversarial attack, and $\Gamma_{\text{train}} := \Gamma(M_{\text{train}})$, $\Gamma_{\text{test}} := \Gamma(M_{\text{test}})$, $\psi_{\text{train}} := \psi(M_{\text{train}})$, and $\psi_{\text{test}} := \psi(M_{\text{test}})$ are functions in Eq. (10).

We further adopt Assumption 2 to help us better understand our robust generalization bound.

Assumption 2. *For adversarial suffix lengths during AT and testing, we assume that $M_{train}, M_{test} \leq \mathcal{O}(N)$, where N is the original ICL prompt length. Besides, for the l_2-norm adversarial perturbation radius, we assume that $\epsilon = \Theta(\sqrt{d})$, where d is the ICL sample dimension.*

In the above Assumption 2, the assumption made on adversarial suffix lengths means that they should not be too long to make the model "forget" the original ICL prompt. Besides, the assumption made on the perturbation radius ϵ ensures that it is large enough to simulate the large (but limited) token vocabulary space of real-world LLMs to help model gain robustness.

Corollary 1. *Suppose Assumption 2 and all conditions in Theorem 2 hold. Suppose $\|\Lambda\|_2 \leq \mathcal{O}(1)$. Then, we have the following robust generalization bound,*

$$\mathcal{R}^{\text{adv}}(\theta_*(M_{\text{train}}), M_{\text{test}}) \leq \mathcal{O}(d) + \mathcal{O}\left(\frac{d^2}{N}\right) + \mathcal{O}\left(N^2 \cdot \frac{M_{\text{test}}^2}{M_{\text{train}}^4}\right).$$

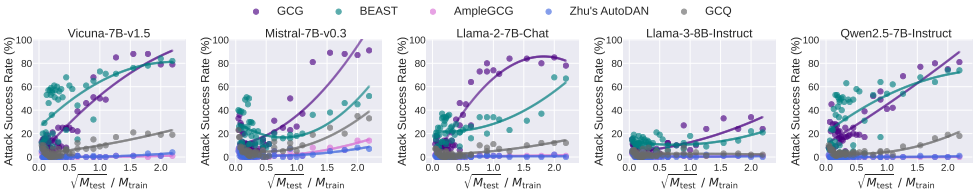

Figure 1: Scatter plots of ASR to the ratio $\sqrt{M_{\text{test}}}/M_{\text{train}}$. For each pair of base model and attack, 48 points are plotted. A high ASR indicates a weak jailbreak robustness.

Corollary 1 is our main theoretical result, which clearly show that for an adversarially trained LSA model, its robust generalization bound depends on the term $\Theta(\sqrt{M_{\text{test}}}/M_{\text{train}})$, where M_{train} and M_{test} are the number of adversarially perturbed in-context samples during training and testing. In other words, for an ICL adversarial attack with an adversarial suffix length $\Theta(M)$, to maintain the order of the robust generalization bound, it is enough to perform surrogate AT with only an adversarial suffix length $\Theta(\sqrt{M})$. Such an observation is useful in practice, since one can thus leverage a "short-length" AT, which requires less GPU memory and training time, to defend against "long-length" jailbreakings.

5 Empirical evidence

In this section, we follow Eq. (3) to perform AT on LLMs and investigate the relationship between adversarial suffix lengths during LLM AT and jailbreak attacks.

5.1 Experimental setup

Models&datasets. We adopt five pre-trained LLMs, which are: Vicuna-7B-v1.5 [70], Mistral-7B-Instruct-v0.3 [21], Llama-2-7B-Chat [52], Llama-3-8B-Instruct [15], and Qwen2.5-7B-Instruct [65]. For data in AT, we use the training set from Harmbench [36] as the safety dataset and Alpaca [50] as the utility dataset. For data in the robustness evaluation, we construct a test set of size 100 that consists of the first 50 samples from the test set of Harmbench [36] and the first 50 samples from AdvBench [73]. For data in the utility analysis, we use the benchmark data from AlpacaEval [10].

Adversarial training. We leverage GCG [73], a token-level jailbreak attack, to synthesize (suffix) jailbreak prompts, in which the adversarial suffix length M_{train} is fixed to one of $\{5, 10, 20, 30, 40, 50\}$ during AT. To reduce computational complexity of tuning LLMs, LoRA [18] is applied to all query and key projection matrices in attentions. In every AT experiment, we follow Eq. (3) to perform AT with Adam. Please refer to Appendix B.2 for omitted settings.

Jailbreak attacks. We use both suffix and non-suffix jailbreak attacks to evaluate the adversarial robustness of trained LLMs. Specifically, five token-level suffix jailbreak attacks are adopted, which are GCG [73], BEAST [44], AmpleGCG [26], Zhu's AutoDAN [71], and GCQ [17]. The adversarial suffix token length M_{test} is varied within $\{5, 10, 20, 40, 60, 80, 100, 120\}$. Meanwhile, two non-suffix jailbreak attacks are leveraged, which are PAIR [7] and DeepInception [25]. Please refer to Appendix B.3 for full implementation details of all used jailbreak attacks.

Evaluations. We focus on evaluating the jailbreak robustness and the utility of trained LLMs. For the robustness evaluation, we report the **Attack Success Rate (ASR)** of jailbreak attacks. An LLM-based judger from [36] is used to determine whether a jailbreak attack succeeds or not. For the utility evaluation, we use the AlpacaEval2 [10] to report the **Length-controlled WinRate (LC-WinRate)** of targeted models against a reference model Davinci003 evaluated under the Llama-3-70B model. An LC-WinRate of 50% means that the output qualities of the two models are equal, while an LC-WinRate of 100% means that the targeted model is consistently better than the reference Davinci003. Please refer to Appendix B.3 for the detailed settings of model evaluations.

5.2 Results analysis

Correlation between the suffix jailbreak robustness and the ratio $\sqrt{M_{\text{test}}}/M_{\text{train}}$. We plot the ASR of models trained and attacked with different adversarial suffix lengths in Figure 1. This results in 48 points for each pair of base model and jailbreak attack. The Pearson correlation coefficient (PCC)

Table 1: PCCs and p-values calculated between ASR and ratio $\sqrt{M_{\text{test}}}/M_{\text{train}}$. A high PCC (within $[-1, 1]$) means a strong correlation between ASR and the ratio. $p < 5.00 \times 10^{-2}$ means that the observation is considered statistically significant.

Model	GCG Attack		BEAST Attack		AmpleGCG Attack		Zhu's AutoDAN		GCQ Attack	
	PCC(↑)	p-value(↓)	PCC(↑)	p-value(↓)	PCC(↑)	p-value(↓)	PCC(↑)	p-value(↓)	PCC(↑)	p-value(↓)
Vicuna-7B	0.93	$\mathbf{4.7 \times 10^{-21}}$	0.63	$\mathbf{1.4 \times 10^{-6}}$	0.19	$\underline{1.9 \times 10^{-1}}$	0.51	$\mathbf{2.5 \times 10^{-4}}$	0.82	$\mathbf{1.4 \times 10^{-12}}$
Mistral-7B	0.86	$\mathbf{4.0 \times 10^{-15}}$	0.29	$\mathbf{4.4 \times 10^{-2}}$	0.74	$\mathbf{1.5 \times 10^{-9}}$	0.49	$\mathbf{3.7 \times 10^{-4}}$	0.70	$\mathbf{2.6 \times 10^{-8}}$
Llama-2-7B	0.88	$\mathbf{9.0 \times 10^{-17}}$	0.67	$\mathbf{1.7 \times 10^{-7}}$	0.37	$\mathbf{1.0 \times 10^{-2}}$	0.13	$\underline{3.8 \times 10^{-1}}$	0.71	$\mathbf{2.1 \times 10^{-8}}$
Llama-3-8B	0.76	$\mathbf{2.8 \times 10^{-10}}$	0.26	$\underline{7.7 \times 10^{-2}}$	-0.07	$\underline{6.2 \times 10^{-1}}$	-0.12	$\underline{4.1 \times 10^{-1}}$	0.0	$\underline{9.7 \times 10^{-1}}$
Qwen2.5-7B	0.87	$\mathbf{1.1 \times 10^{-15}}$	0.58	$\mathbf{1.0 \times 10^{-5}}$	-0.24	$\underline{1.0 \times 10^{-1}}$	0.16	$\underline{2.6 \times 10^{-1}}$	0.72	$\mathbf{1.1 \times 10^{-8}}$

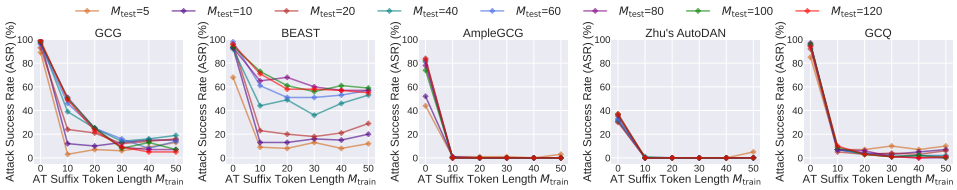

Figure 2: ASR versus M_{train} on Vicuna-7B-v1.5 under jailbreaking with different M_{test}. $M_{\text{train}} = 0$ means that AT is not performed on the evaluated model. A low ASR indicates a strong robustness.

and the corresponding p-value between the ratio $\sqrt{M_{\text{test}}}/M_{\text{train}}$ and the ASR are calculated in Table 1, where **bold p-values** indicate that observations are statistically significant (*i.e.*, $p < 0.05$), while underlined ones indicate they are not significant.

When the jailbreak attack used during AT is the same as that used during robustness evaluation (*i.e.*, GCG), one can observe from Figure 1 that a clear positive correlation between the ratio $\sqrt{M_{\text{test}}}/M_{\text{train}}$ and the ASR for all evaluated base models. Further, high PCCs (> 0.7) and low p-values (< 0.05) in Table 1 also confirm that the observed positive correlation is statistically significant.

Besides, when the jailbreak attack is BEAST and GCQ, which is different from that used during AT, the significant positive correlation between the ratio $\sqrt{M_{\text{test}}}/M_{\text{train}}$ and the ASR can only be observed from some of the base models. This may be due to the fact that AT with only a single jailbreak attack may not help the model generalize well to unseen attacks. Therefore, it might be necessary to adopt multiple attacks when performing AT-based alignment on LLMs. Nevertheless, from Figure 1, we find that for those models where the correlation is not significant (*i.e.*, Mistral-7B, and Llama-3-8B), GCG-based AT can still suppress the ASR to no more than 50%, which indicates that it can still help models gain a certain degree of robustness against unseen attacks.

Finally, for AmpleGCG and Zhu's AutoDAN attacks, we notice that the correlation between the ratio $\sqrt{M_{\text{test}}}/M_{\text{train}}$ and the ASR cannot be observed on most of the base models. However, this is simply due to AT being too effective in defending against these two attacks: from Figure 1, one can observe that AT effectively reduces ASRs of AmpleGCG and Zhu's AutoDAN to nearly zero in most cases.

Relationship between adversarial suffix lengths in AT (*i.e.*, M_{train}) and suffix jailbreaking (*i.e.*, M_{test}). We plot curves of the ASR on Vicuna-7B versus the adversarial suffix token length during AT in Figure 2. Results on remaining base models are presented in Figure 4 in Appendix B.4. From these figures, we find that as the adversarial suffix token length during AT increases, AT can effectively reduce the ASR of all analyzed attacks. Furthermore, when the AT adversarial suffix token length is set to 20, AT is already able to reduce the ASR by at least 30% under all settings. All these results demonstrate the effectiveness of defending against long-length jailbreaking with short-length AT.

Table 2: Time cost (hrs) of LLM AT with different adversarial suffix lengths.

Model	Adversarial Suffix Token Length M_{train} in AT					
	5	10	20	30	40	50
Vicuna-7B	10.2h	11.3h	13.8h	16.0h	18.2h	20.4h
Mistral-7B	8.9h	9.9h	12.0h	14.3h	16.6h	19.0h
Llama-2-7B	9.9h	11.0h	13.2h	15.5h	18.1h	20.0h
Llama-3-8B	9.7h	10.8h	13.1h	15.3h	17.7h	20.2h
Qwen2.5-7B	9.1h	9.9h	11.7h	13.9h	16.4h	18.4h

Time cost of LLM AT with different adversarial suffix lengths M_{train}. We then present the time costs of performing LLM AT in Table 2. From the table, we find that when the adversarial suffix